郁然, 未央-水木22 数值分析

1. 证: $x^{(i)}$ 正交

设 $v^{(0)} = \sum_{j=1}^{n} \alpha_j \overrightarrow{x^{(j)}}$

由题意: $\alpha_1 \neq 0$

① 若 $\alpha_2 = \alpha_3 = \cdots = \alpha_n = 0$

∴ $v^{(0)} = \alpha_1 \vec{x}^{(1)}$

~~$v^{(k+1)} = Av^{(k)}$~~

$$v^{(1)} = \frac{\alpha_1 \lambda_1 \vec{x}^{(1)}}{\|\alpha_1 \lambda_1 \vec{x}^{(1)}\|_2}$$

$$R_1 = \frac{\alpha_1^2 \lambda_1^3 \|\vec{x}^{(1)}\|_2^2}{\alpha_1^2 \lambda_1^2 \|\vec{x}^{(1)}\|_2^2} = \lambda_1$$

~~此时式不成立，C不存在~~ 此时显然成立，我们可取 $C=1$

② $\alpha_2, \ldots, \alpha_n$ 不全为0 不妨 $\alpha_s \neq 0$ $s>1$ 且对 $\forall\ 1<t<s$, $\alpha_t = 0$

$$v^{(1)} = \frac{\sum_{j=1}^{n} \alpha_j \lambda_j \vec{x}^{(j)}}{\|\sum_{j=1}^{n} \alpha_j \lambda_j \vec{x}^{(j)}\|_2}$$

$$R_0 = \frac{\sum_{j=1}^{n} \alpha_j^2 \lambda_j \|\vec{x}^{(j)}\|_2^2}{\sum_{j=1}^{n} \alpha_j^2 \|x^{(j)}\|_2^2}$$

$$R_1 = \frac{\sum_{j=1}^{n} \alpha_j^2 \lambda_j^3 \|\vec{x}^{(j)}\|_2^2}{\sum_{j=1}^{n} \alpha_j^2 \lambda_j^2 \|x^{(j)}\|_2^2}$$

$$v^{(k)} = \frac{\sum_{j=1}^{n} \alpha_j \lambda_j^k \vec{x}^{(j)}}{\|\sum_{j=1}^{n} \alpha_j \lambda_j^k \vec{x}^{(j)}\|_2}$$

$$R_k = \frac{\sum_{j=1}^{n} \alpha_j^2 \lambda_j^{2k+1} \|\vec{x}^{(j)}\|_2^2}{\sum_{j=1}^{n} \alpha_j^2 \lambda_j^{2k} \|\vec{x}^{(j)}\|_2^2}$$

$$|R_k - \lambda_1| = \left| \frac{\sum_{j=2}^{n} \alpha_j^2 \lambda_j^{2k} \|\vec{x}^{(j)}\|_2^2 (\lambda_j - \lambda_1)}{\sum_{j=1}^{n} \alpha_j^2 \lambda_j^{2k} \|\vec{x}^{(j)}\|_2^2} \right|$$

$$\le \frac{\left|\sum_{j=2}^{n} \alpha_j^2 \lambda_j^{2k} \|\vec{x}^{(j)}\|_2^2 (\lambda_j - \lambda_1)\right|}{\alpha_1^2 \lambda_1^{2k} \|\vec{x}^{(1)}\|_2^2} = \frac{\sum_{j=2}^{n} \alpha_j^2 \lambda_j^{2k} \|\vec{x}^{(j)}\|_2^2 (\lambda_1 - \lambda_j)}{\alpha_1^2 \lambda_1^{2k} \|\vec{x}^{(1)}\|_2^2}$$

$$\le \frac{(n-s+1)\alpha_s^2 \lambda_s^{2k} \max_{1\le j\le n} \|\vec{x}^{(j)}\|_2^2 \cancel{(\lambda_1-\lambda_n)} \max_{1\le j\le n}(\lambda_1 - \lambda_j)}{\alpha_1^2 \lambda_1^{2k} \|\vec{x}^{(1)}\|_2^2}$$

$$\le \frac{(n-s+1)\alpha_s^2 \max_{1\le j\le n} \|\vec{x}^{(j)}\|_2^2 \max_{1\le j\le n}(\lambda_1 - \lambda_j)}{\alpha_1^2 \|\vec{x}^{(1)}\|_2^2} \left|\frac{\lambda_2}{\lambda_1}\right|^{2k}$$

→ 取为C即可

扫描全能王 创建

2. (1) $H\vec{b} = c\vec{e_1}$

设 $\vec{x_1}+\vec{x_2}=\vec{b}$

$\vec{x_1}-\vec{x_2}=c\vec{e_1}$

$$\vec{x_2}=\frac{\vec{b}-c\vec{e_1}}{2}$$

$\vec{v}$ 与 $\vec{x_2}$ 同向

$$\therefore \vec{v}=\frac{\vec{b}-c\vec{e_1}}{\|\vec{b}-c\vec{e_1}\|_2}=\frac{1}{\sqrt{(1-c)^2+2}}\begin{pmatrix}1-c\\1\\1\end{pmatrix}$$

$$H=I-2v^{*}v^{*}=I-\frac{2}{(1-c)^2+2}\begin{pmatrix}(1-c)^2 & (1-c) & (1-c)\\ 1-c & 1 & 1\\ 1-c & 1 & 1\end{pmatrix}$$

$$=\begin{pmatrix}1-\frac{2(1-c)^2}{(1-c)^2+2} & -\frac{2(1-c)}{(1-c)^2+2} & -\frac{2(1-c)}{(1-c)^2+2}\\ -\frac{2(1-c)}{(1-c)^2+2} & \frac{(1-c)^2}{(1-c)^2+2} & -\frac{2}{(1-c)^2+2}\\ -\frac{2(1-c)}{(1-c)^2+2} & -\frac{2}{(1-c)^2+2} & \frac{(1-c)^2}{(1-c)^2+2}\end{pmatrix}$$

$c=\sqrt{3}$ 或 $-\sqrt{3}$

(2) ~~[illegible]~~

~~[illegible]~~

~~不存在这样的x~~

$$\begin{pmatrix}\cos\theta_1 & & \sin\theta_1\\ & 1 & \\ -\sin\theta_1 & & \cos\theta_1\end{pmatrix}\vec{b}=\begin{pmatrix}x_1\\x_2\\0\end{pmatrix}$$

$\theta_1=\frac{\pi}{4}$ $x_1=\sqrt{2}$ $x_2=1$

$$\begin{pmatrix}\cos\theta_2 & \sin\theta_2 & \\ -\sin\theta_2 & \cos\theta_2 & \\ & & 1\end{pmatrix}\begin{pmatrix}\sqrt{2}\\1\\0\end{pmatrix}=\begin{pmatrix}c\\0\\0\end{pmatrix}$$

$c=\pm\sqrt{3}$

$c=\sqrt{3}$ 时

$\underset{\sin}{\cancel{\cos}}\theta_2=\frac{\sqrt{3}}{3}$ $\cancel{\sin\theta_2=}$ $\cos\theta_2=\frac{\sqrt{6}}{3}$

$$此时\ J=\begin{pmatrix}\frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & \\ -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & \\ & & 1\end{pmatrix}\begin{pmatrix}\frac{\sqrt{2}}{2} & & \frac{\sqrt{2}}{2}\\ & 1 & \\ -\frac{\sqrt{2}}{2} & & \frac{\sqrt{2}}{2}\end{pmatrix}=\begin{pmatrix}\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3}\\ -\frac{\sqrt{6}}{6} & \frac{\sqrt{6}}{3} & -\frac{\sqrt{6}}{6}\\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2}\end{pmatrix}$$

$c=-\sqrt{3}$ 时

$\sin\theta_2=-\frac{\sqrt{3}}{3}$ $\cos\theta_2=-\frac{\sqrt{6}}{3}$

$$此时\ J=\begin{pmatrix}-\frac{\sqrt{6}}{3} & -\frac{\sqrt{3}}{3} & \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{3} & \\ & & 1\end{pmatrix}\begin{pmatrix}\frac{\sqrt{2}}{2} & & \frac{\sqrt{2}}{2}\\ & 1 & \\ -\frac{\sqrt{2}}{2} & & \frac{\sqrt{2}}{2}\end{pmatrix}=\begin{pmatrix}-\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3}\\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{6}}{3} & \frac{\sqrt{6}}{6}\\ -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2}\end{pmatrix}$$

扫描全能王 创建

郁然， 未央-水木22 数值分析

3. 证：设 $A=(\vec{a_1},\cdots,\vec{a_n})$

其中 $\vec{a_i}=\begin{pmatrix}a_{1i}\\ \vdots\\ a_{ii}\\ a_{i+1,i}\\ 0\\ \vdots\\ 0\end{pmatrix}$ $(i\neq n)$ 且 $a_{1i},\cdots,a_{i+1,i}$ 均不为0 $\quad \vec{a_n}=\begin{pmatrix}a_{1n}\\ \vdots\\ a_{nn}\end{pmatrix}$ $\quad a_{1n},\cdots,a_{nn}$ 均不为0

设 $Q=(\vec{q_1},\cdots,\vec{q_n})$ $\quad R=\begin{pmatrix}r_{11} & \cdots & r_{1n}\\ 0 & \ddots & \vdots\\ & & r_{nn}\end{pmatrix}$

$$A=QR=(\vec{q_1},\cdots,\vec{q_n})\begin{pmatrix}r_{11} & \cdots & r_{1n}\\ 0 & \ddots & \vdots\\ & & r_{nn}\end{pmatrix}$$

$$\therefore \vec{a_i}=\sum_{j=1}^{i}\vec{q_j}\cdot r_{ji}$$

$\therefore \vec{q_j}=\begin{pmatrix}q_{1j}\\ \vdots\\ q_{jj}\\ q_{j+1,j}\\ 0\\ \vdots\\ 0\end{pmatrix}$ $(j\neq n)$ 且 $q_{1j},\cdots,q_{j+1,j}$ 均不为0

$\vec{q_n}=\begin{pmatrix}q_{1n}\\ \vdots\\ q_{nn}\end{pmatrix}$ $\quad q_{1n},\cdots,q_{nn}$ 可以为0

$$B=RQ=\begin{pmatrix}r_{11} & \cdots & r_{1n}\\ 0 & \ddots & \vdots\\ & & r_{nn}\end{pmatrix}\begin{pmatrix}q_{11} & \cdots & q_{1,n-1} & q_{1n}\\ q_{21} & & \vdots & \vdots\\ & \ddots & q_{n,n-1} & q_{nn}\end{pmatrix}=\begin{pmatrix}b_{11} & \cdots & b_{1n}\\ b_{21} & \cdots & b_{2n}\\ \vdots & \ddots & \vdots\\ b_{n1} & \cdots & b_{nn}\end{pmatrix}$$

对于 b_{ij} $(i>j+1,$ 且 $j\leq n-2,\ i\leq n)$

计算可知 $b_{ij}=0$

$\therefore$ B 仍然是同形式的上 Hessenberg 矩阵

未央实践

扫描全能王 创建